

Analysis PhD Exam, January 1989

Answer 6 out of the 7 problems. Justify all work. To obtain any partial credit, all work must be presented in a neat, logical and concise fashion.

1. Let f be a Borel-measurable function defined on $[0, 1]$ with $\infty > f(x) \geq 0$.

Part (a) Prove that there is a non-increasing function f^* defined on $[0, 1]$ such that

$$m\{x : f^*(x) \geq \lambda\} = m\{x : f(x) \geq \lambda\}$$

for every $\lambda \geq 0$, where m is Lebesgue measure on $[0, 1]$. (f^* is called a monotone rearrangement of f .)

Part (b) Show that $\int f^* dm = \int f dm$.

2. Let $(X, \mathcal{F}, \mu)$ be a finite measure space, and let f and g be two measurable functions such that

$$\mu(f \in A, g \in B) = \mu(f \in A)\mu(g \in B)$$

for every $A, B \in \mathcal{B}(\mathbb{R})$. Let $\nu(A) = \mu(g \in A)$. Show that

$$\int H(f, g) d\mu = \int \int H(f, t) \nu(dt) d\mu$$

for every positive $\mathcal{B}(\mathbb{R}^2)$ -measurable function H .

3. Consider $\ell^\infty = \{(a_n)_{n=1}^\infty : \sup_n |a_n| < \infty\}$. This space becomes a Banach space when equipped with the norm $\|(a_n)_{n=1}^\infty\| = \sup\{|a_n| : n \geq 1\}$.

Part (a) Show there is a linear functional $T : \ell^\infty \rightarrow \mathbb{R}$ with the property: if $\lim_{n \rightarrow \infty} a_n$ exists, then

$$T[(a_n)_{n=1}^\infty] = \lim_{n \rightarrow \infty} a_n.$$

(Use the Hahn–Banach theorem.)

Part (b) Modify your proof in (a) to show there is a linear functional $S : \ell^\infty \rightarrow \mathbb{R}$ with the property: if $\lim_{n \rightarrow \infty} n^{-1} \sum_{k \leq n} a_k$ exists, then

$$S[(a_n)_{n=1}^\infty] = \lim_{n \rightarrow \infty} n^{-1} \sum_{k \leq n} a_k.$$

4. State and prove the Radon–Nikodym theorem.
5. State and prove Fubini's theorem.
6. Let μ be a measure on $((0, \infty), \mathcal{B}(0, \infty))$ defined by

$$\mu(A) = \int_A |x|^{-1/2} dx.$$

Let $T : (0, \infty) \rightarrow (0, \infty)$ be given by $T(x) = x^{-1}$. Define a new measure π on $(0, \infty)$ by setting $\pi(A) = \mu(T^{-1}(A))$. Compute $d\pi/d\mu$.

7. Let $c(t)$ be a function defined on $[0, \infty)$ with the following properties: $c(t)$ is strictly increasing and continuous, $c(0) = 0$, and $c(\infty) = \infty$. Let $\tau_t = \inf\{s : c(s) > t\}$. Prove that

$$\int_0^\infty f(t) c(dt) = \int_0^\infty f(\tau_t) dt$$

for every positive continuous function f on $[0, \infty)$.